

Seaborn Bar Plot (sns.barplot) Notes

1. What is a Bar Plot?

A bar plot compares an aggregated numerical value (mean by default) across categories.

2. When to Use

Compare average salary by department, average sales by month, marks by class, etc.

3. Rules

- One categorical variable and one numerical variable.
- Default estimator is mean.
- Use estimator to change the statistic.
- Use strip/swarm plots to show every observation.

4. Basic Syntax

```
sns.barplot(x='category', y='value', data=df)
```

5. Important Parameters

x: categorical variable.

y: numerical variable.

data: DataFrame source.

hue: compare another categorical group.

palette: change colors.

estimator: choose statistic (mean, median, max, min, sum).

errorbar: show variability/uncertainty.

capsize: add caps to error bars.

width: control bar width.

dodge: separate hue groups side by side.

order: custom category order.

orient: horizontal/vertical orientation.

alpha: transparency.

6. Common Mistakes

Don't use barplot for counts (use countplot). Don't use two numerical variables.

7. Bar Plot vs Count Plot

Bar plot summarizes a numerical variable; count plot shows category frequencies.

8. Interview Points

Default estimator = mean. Hue compares groups. Estimator changes statistic. Errorbar shows variability.

9. Quick Revision

Purpose: Compare aggregated numerical values across categories. Required: one categorical + one numerical variable.